

ANIMAL HABITATS

Two-thirds of Earth's surface is covered by oceans. It is this abundance of water that enables the planet to support billions of living organisms, both in the seas and on the continents and islands that make up the land. The environment in which an organism lives is its habitat, and the huge range of habitats found on land are home to a vast number of plant species and a spectacular diversity of animals.

Geographers divide the world into ecoregions characterized by broad habitat types such as forest, grassland, wetland, desert, or polar zones. These can in turn be subdivided almost endlessly into more precise habitats, each supporting a unique community of plant and animals.

Climate exerts a powerful influence over each of the world's great ecoregions. Energy input from the Sun is greatest at the tropics, and the transfer of this energy via the atmosphere and oceans generates the currents of air and water that drive the world's weather systems. On land, weathering of rock leads to the formation of soil in which plants grow, forming the basis of ecological communities.

Variety of life

Biodiversity refers to the variety of life in a given habitat or ecoregion. As a rule, biodiversity increases toward the equator, with tropical forests and warm coastal seas registering the greatest numbers of species. Both poles are inhabited by comparatively few animal species, but in the Arctic, many animals live on land whereas in Antarctica, most animal life is found in the ocean.

Plants and animals become better adapted to their particular habitat through natural selection—those most suited to the environmental conditions survive in greater numbers and produce more offspring. This is a continuous process as habitats change slowly over time. Sudden events such as volcanic eruptions, floods, or human development can have catastrophic impacts, especially on species that are specialized to a certain way of life. So-called generalist species cope better with change, such as a fall or rise in temperature, but may be displaced by specialist species when conditions stabilize.

ECOREGION

The Alps comprise one of the world's best-known montane ecoregions. They extend into eight European countries, forming a snow-tipped arc that stretches from France and Italy in the southwest to Austria in the east.

CONTINENT

In southern Europe, the warm waters of the Mediterranean lap the coast and much of the land is covered by Mediterranean woodland and scrub. The soaring mountains of the Alps form a physical barrier, beyond which lies the colder, wetter north.



THE ALPS ECOREGION

If you pick any place on the planet, you will find a unique set of conditions, influenced by geography, latitude, and climate. This means the area, such as the Alps, will be home to a unique community of plants and animals.

Scientists estimate that only **14 percent of species** have been identified so far, and of those, **91 percent live on land**